CS111 Code Cheatsheet: Processes, Exec, Pipes, FDs

API Shape: Memorize Returns

call	return / meaning	trap
`fork()`	child sees `0`; parent sees child PID; `-1` error	both continue after call
`execvp(file,argv)`	returns only on failure	`argv` must end with `NULL`
`waitpid(pid,&st,0)`	wait/reap one child	`pid=-1` means any child
`pipe(p)`	`p[0]` read, `p[1]` write	EOF only when all write ends closed
`dup2(old,new)`	makes `new` refer to old target	direction matters

Fully Annotated Shell Command

```
#include <sys/wait.h>
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>

int main(void) {
  char *argv[] = {"wc", "-l", "in.txt", NULL};
  pid_t pid = fork();

  if (pid < 0) { perror("fork"); exit(1); }

  if (pid == 0) {
    // Child only: replace child with wc.
    execvp(argv[0], argv);
    // Only reached if execvp failed.
    perror("execvp");
    exit(1);
  }

  // Parent only: wait keeps child from becoming zombie.
  int st;
  waitpid(pid, &st, 0);
  if (WIFEXITED(st)) {
    printf("exit=%d\n", WEXITSTATUS(st));
  }
}
```

- `execvp` keeps same PID but replaces code/heap/stack/globals.
- Parent does not run `wc`; child does.
- `perror/exit` after `execvp` is failure path.

Multiple Fork Trace

```
pid_t a = fork();
pid_t b = fork();
printf("a=%d b=%d\n", a, b);
```

process	a	b	why
P0	PID(P1)	PID(P2)	parent of both forks
P1	0	PID(P3)	child of first, parent of second
P2	PID(P1)	0	child of second from P0; copied `a`
P3	0	0	child of second from P1

Rule: `0` only in the newly-created child of that exact `fork`.

Conditional Forks: Count Processes

```
if (fork() == 0) {
  fork();
}
printf("x\n");
```

who	first fork	second fork?	prints
original parent	nonzero	no	yes
first child	0	yes, parent side	yes
grandchild	copied 0	created by second	yes

Answer: three prints. Only processes that see first result `0` enter the body.

Child Loop: Correct Pattern

```
for (int i = 0; i < 3; i++) {
  pid_t pid = fork();
  if (pid < 0) exit(1);

  if (pid == 0) {
    do_child_work(i);
    exit(0);   // without this, child keeps forking
  }
}

while (waitpid(-1, NULL, 0) > 0) {
  // reap all children
}
```

Exam smell: "create N children." Child must usually stop after its assigned work.

Wait Status: Which Child Failed?

```
for (int i = 0; i < n; i++) {
  pid_t pid = fork();
  if (pid == 0) {
    exit(run_job(i));        // child returns status 0..255
  }
  pids[i] = pid;
}

for (int i = 0; i < n; i++) {
  int st;
  pid_t done = waitpid(pids[i], &st, 0);
  if (done < 0) perror("waitpid");
  if (WIFEXITED(st) && WEXITSTATUS(st) != 0) {
    printf("job %d failed\n", i);
  }
}
```

`waitpid(pids[i],...)` waits for a specific child; `waitpid(-1,...)` waits for whichever finishes next.

Redirect Child Stdout To File

```
int fd = open("out.txt", O_WRONLY|O_CREAT|O_TRUNC, 0644);
pid_t pid = fork();

if (pid == 0) {
  dup2(fd, STDOUT_FILENO); // fd 1 now points to out.txt
  close(fd);               // no longer need original number
  char *args[] = {"echo", "hi", NULL};
  execvp(args[0], args);
  perror("execvp");
  exit(1);
}

close(fd);                 // parent should close too
waitpid(pid, NULL, 0);
```

Wrong direction: `dup2(STDOUT_FILENO, fd)` does not redirect stdout.

Use All Three Standard FDs

```
char buf[128];
int n = read(STDIN_FILENO, buf, sizeof(buf));
if (n < 0) {
  write(STDERR_FILENO, "read failed\n", 12);
  exit(1);
}
write(STDOUT_FILENO, buf, n);
```

- `stdin=0`: input source.
- `stdout=1`: normal output.
- `stderr=2`: error output, often not redirected with stdout.

Pipeline: `left | right`

```
int p[2];
pipe(p);

pid_t left_pid = fork();
if (left_pid == 0) {
  dup2(p[1], STDOUT_FILENO); // left writes into pipe
  close(p[0]); close(p[1]);
  execvp(left[0], left);
  perror("left"); exit(1);
}

pid_t right_pid = fork();
if (right_pid == 0) {
  dup2(p[0], STDIN_FILENO);  // right reads from pipe
  close(p[0]); close(p[1]);
  execvp(right[0], right);
  perror("right"); exit(1);
}

// Parent neither reads nor writes pipeline data.
close(p[0]);
close(p[1]);
waitpid(left_pid, NULL, 0);
waitpid(right_pid, NULL, 0);
```

- If parent keeps `p[1]`, right side may never see EOF.
- Close unused ends after `dup2` in every process.

Pipe EOF Bug: Why It Hangs

```
pipe(p);
if (fork() == 0) {
  close(p[1]);
  while (read(p[0], buf, sizeof(buf)) > 0) {}
  exit(0);
}

// BUG: parent writes but forgets close(p[1])
write(p[1], "abc", 3);
waitpid(-1, NULL, 0); // child waits forever for EOF
```

Reader sees EOF only after every copy of the write end is closed in every process.

Command-Line Arguments

```
int main(int argc, char *argv[]) {
  // argv[0] is program name
  // argv[argc] is always NULL
  for (int i = 1; i < argc; i++) {
    printf("arg %d = %s\n", i, argv[i]);
  }
}
```

`argv` is an array of `char *`. Each string is a separate null-terminated char array.

Pipe Without Exec: Direction Sanity

```
pipe(p);
if (fork() == 0) {
  close(p[1]);             // child reads only
  char buf[100];
  int n = read(p[0], buf, 99);
  buf[n] = '\0';
  printf("%s\n", buf);
  exit(0);
}

close(p[0]);               // parent writes only
write(p[1], "hello", 5);
close(p[1]);               // sends EOF
waitpid(-1, NULL, 0);
```

Use this mental model before adding redirection and `execvp`.

FD Sharing After Fork

```
int fd = open("data", O_RDONLY);
if (fork() == 0) {
  char c; read(fd, &c, 1);
  printf("child %c\n", c);
  exit(0);
}
char c; read(fd, &c, 1);
printf("parent %c\n", c);
waitpid(-1, NULL, 0);
```

- FD table entries are copied.
- They may point to same open-file description.
- Open-file description stores current file offset.
- Scheduling decides who consumes first byte.

Build `argv` For `execvp`

```
char *cmd[] = {
  "grep",       // argv[0]: conventionally program name
  "-n",
  "needle",
  "file.txt",
  NULL          // required terminator
};
execvp(cmd[0], cmd);
```

`execvp` searches `PATH` for `cmd[0]` if it has no slash. With `./prog` or `/bin/ls`, it uses that path.

Monitor API Pattern

```
std::mutex m;
std::condition_variable cv;

void method() {
  std::unique_lock<std::mutex> lock(m);
  while (!predicate_over_shared_state) {
    cv.wait(lock);  // unlocks while sleeping; relocks before return
  }
  // update shared state while lock is held
  cv.notify_all();  // or notify_one if one waiter is enough
}
```

CV has no memory. Predicate is the truth; CV only wakes threads to re-check it.

Final Monitor Pattern Map

exam shape	state	wake
CalTrain	waiting, seats, boarding	passengers + train CVs
2022 Merge	waiting per lane, total, current merge	notify one chosen lane
2023 Flight	sorted map position -> waiter CV	notify lowest key
2024 Inventory	counts per stock id	notify all, re-check whole order
Party/Dorm	bucket -> queue of waiter records	match opposite/same bucket
H2O/exact group	queues of waiter records	mark exact records assigned

In every monitor: lock, update waiting state before sleep, `while` wait, change state before notify, return with invariant true.

Lost Update + Fix

```
int x = 0;
void bad_worker() {
  x++;        // load, add, store: not atomic
}
```

```
// Bad interleaving:
T1 reads x=0
T2 reads x=0
T1 writes x=1
T2 writes x=1   // one increment lost
```

```
std::mutex m;
void good_worker() {
  std::unique_lock<std::mutex> lock(m);
  x++;
} // auto-unlock
```

Protect the whole read-modify-write, not just the write.

Create Threads And Join

```
void worker(int id) {
  do_work(id);
}

int main() {
  std::vector<std::thread> threads;
  for (int i = 0; i < 4; i++) {
    threads.push_back(std::thread(worker, i));
  }

  for (std::thread &t : threads) {
    t.join();       // wait for that thread to finish
  }
}
```

`join` is thread-world `waitpid`: it waits for completion and cleans up thread resources.

Bounded Buffer

```
class BoundedBuffer {
  std::mutex m;
  std::condition_variable nonempty, nonfull;
  std::queue<int> q;
  int cap;
public:
  void put(int x) {
    std::unique_lock<std::mutex> lock(m);
    while ((int)q.size() == cap) {
      nonfull.wait(lock);
    }
    q.push(x);                  // state change
    nonempty.notify_one();      // a getter may proceed
  }

  int get() {
    std::unique_lock<std::mutex> lock(m);
    while (q.empty()) {
      nonempty.wait(lock);
    }
    int x = q.front();
    q.pop();                    // state change
    nonfull.notify_one();       // a putter may proceed
    return x;
  }
};
```

Two predicates means two CVs are often clearer.

CalTrain Worked Pattern

```
class Station {
  mutex m; condition_variable train, all_boarded;
  int waiting=0, seats=0, boarding=0;
public:
  void load_train(int available) {
    unique_lock<mutex> lock(m);
    seats = available;
    train.notify_all();
    while ((seats > 0 && waiting > 0) || boarding > 0) {
      all_boarded.wait(lock);
    }
    seats = 0;
  }

  void wait_for_train() {
    unique_lock<mutex> lock(m);
    waiting++;
    while (seats == 0) train.wait(lock);
    waiting--; seats--; boarding++;
  } // unlock while passenger physically boards

  void boarded() {
    unique_lock<mutex> lock(m);
    boarding--;
    if ((seats == 0 || waiting == 0) && boarding == 0)
      all_boarded.notify_one();
  }
};
```

Trap: train cannot leave just because seats hit 0; passengers allowed to board still must call `boarded()`.

Bridge / RW Predicate Mini-Patterns

```
// bridge direction fairness
while (crossing_opp > 0 ||
       (waiting_opp > 0 && consecutive_mine >= 5))
  cv_mine.wait(lock);
waiting_mine--; crossing_mine++; consecutive_mine++;
consecutive_opp = 0;

// writer-preference readers/writers
while (writers > 0 || waiting_writers > 0) cv.wait(lock);
readers++;
while (readers > 0 || writers > 0) cv.wait(lock);
writers = 1;
```

These are predicate questions: identify the exact Boolean that makes entry safe/fair.

2022 Merge: Fair Category Rotation

```
class Merge {
  mutex m;
  vector<int> waiting;
  vector<condition_variable> safe;
  int total=0, prev=0;
  bool merging=false;
public:
  void wait(int lane) {
    unique_lock<mutex> lock(m);
    if (!merging) { merging=true; return; }
    waiting[lane]++; total++;
    safe[lane].wait(lock);    // selected lane only
  }

  void merged() {
    unique_lock<mutex> lock(m);
    if (total == 0) { merging=false; return; }
    do { prev = (prev + 1) % waiting.size(); }
    while (waiting[prev] == 0);
    waiting[prev]--; total--;
    safe[prev].notify_one();  // exactly one car
  }
};
```

Pattern: categories + fairness. Keep `merging=true` when handing directly to next car.

2023 Flight: Ordered Waiters

```
class Flight {
  mutex m;
  map<size_t, condition_variable*> ready;
public:
  void wait_to_board(size_t pos) {
    unique_lock<mutex> lock(m);
    condition_variable mine;
    ready.emplace(pos, &mine);
    mine.wait(lock);          // board_next selects me
  }

  bool board_next() {
    unique_lock<mutex> lock(m);
    if (ready.empty()) return false;
    auto it = ready.begin();  // lowest waiting position
    it->second->notify_one();
    ready.erase(it);
    return true;
  }
};
```

`map` sorted order solves “lowest present position first; skipped late passenger can still beat higher waiting positions.”

Exact Selected Waiters: H2O Shape

```
struct Waiter {
  bool assigned = false;
  std::condition_variable cv;
};

std::mutex m;
std::deque<Waiter*> H, O;

void try_group() {
  if (H.size() >= 2 && O.size() >= 1) {
    Waiter *h1 = H.front(); H.pop_front();
    Waiter *h2 = H.front(); H.pop_front();
    Waiter *o  = O.front(); O.pop_front();
    h1->assigned = true;
    h2->assigned = true;
    o->assigned = true;
    h1->cv.notify_one();
    h2->cv.notify_one();
    o->cv.notify_one();
  }
}

void hydrogen() {
  Waiter self;
  std::unique_lock<std::mutex> lock(m);
  H.push_back(&self);
  try_group();
  while (!self.assigned) self.cv.wait(lock);
  lock.unlock();
  bond();
}
```

- `h1->cv` means `(*h1).cv`.
- Use identity records when prompt says "exactly selected threads."

Party / Dorm Matching

```
struct Person {
  string name, result;
  bool matched=false;
  condition_variable cv;
};

mutex m;
map<Key, deque<Person*>> waiting;

string meet(string name, Key mine, Key want) {
  unique_lock<mutex> lock(m);
  Key opposite = {want, mine};
  auto &q = waiting[opposite];
  if (!q.empty()) {
    Person *other = q.front(); q.pop_front();
    other->result = name;
    other->matched = true;
    other->cv.notify_one();
    return other->name;
  }

  Person self; self.name = name;
  waiting[{mine,want}].push_back(&self);
  while (!self.matched) self.cv.wait(lock);
  return self.result;
}
```

Dorm variant: key is dorm id and opposite bucket is same bucket. Use queue for FIFO within bucket.

All-Or-Nothing Inventory

```
std::mutex m;
std::condition_variable changed;
std::map<int,int> stock;

bool can_fill(vector<int> ids, vector<int> counts) {
  for (int i = 0; i < ids.size(); i++) {
    if (stock[ids[i]] < counts[i]) return false;
  }
  return true;
}

void order(vector<int> ids, vector<int> counts) {
  std::unique_lock<std::mutex> lock(m);
  while (!can_fill(ids, counts)) {
    changed.wait(lock);
  }
  // subtract only after whole order is possible
  for (int i = 0; i < ids.size(); i++) {
    stock[ids[i]] -= counts[i];
  }
}

void add(int id, int n) {
  std::unique_lock<std::mutex> lock(m);
  stock[id] += n;
  changed.notify_all(); // different orders need different items
}
```

Wrong: consume item A, then sleep waiting for item B. That creates partial reservation bugs.

Last-Section Wrong-Answer Traps

```
CV wait:     while (!predicate) cv.wait(lock);  // not if
Inventory:  check all resources, then subtract all
Flight:     map.begin() gives lowest waiting key
Merge:      notify_one selected lane; do not notify_all cars
Party/Dorm: pop waiter before notify; local Waiter is safe while blocked
CalTrain:   count boarding passengers, not just available seats
Deadlock:   every thread must acquire locks in same global order
```

`notify_one`: selected waiter/category. `notify_all`: many possible predicates, each re-checks.

CS111 Code Cheatsheet: Unix V6 Filesystem Final Patterns

V6 Constants / Struct Reminders

```
#define DISKIMG_SECTOR_SIZE 512
#define INODE_START_SECTOR  2
#define ROOT_INUMBER        1
#define MAX_COMPONENT_LENGTH 14

// direntv6: 2-byte inode + 14-byte name
// 512 / 16 = 32 dirents per block
// uint16_t block ptrs: 512 / 2 = 256 ptrs
```

- `IALLOC`: inode allocated.
- `IFMT`: file type mask. `IFDIR`: directory.
- `ILARG`: large V6 inode addressing.
- `size = (i_size0 << 16) | i_size1`.

`inode_iget`: Inumber To Inode

```
int inode_iget(fs, int inumber, struct inode *inp) {
  struct inode inodes[DISKIMG_SECTOR_SIZE /
                      sizeof(struct inode)];

  int idx = inumber - 1; // inumber 1 is array index 0
  int per = DISKIMG_SECTOR_SIZE / sizeof(struct inode);
  int sector = INODE_START_SECTOR + idx / per;
  int offset = idx % per;

  if (diskimg_readsector(fs->dfd, sector, inodes)
      != DISKIMG_SECTOR_SIZE) {
    return -1;
  }
  *inp = inodes[offset];
  return 0;
}
```

Disk layout: sector 0 boot, sector 1 superblock, sector 2 starts inode list.

V6 `inode_indexlookup`

```
int inode_indexlookup(fs, struct inode *in, int b) {
  uint16_t ptrs[256], ptrs2[256];

  if (!(in->i_mode & ILARG)) {
    return in->i_addr[b]; // small: direct sectors
  }

  if (b < 7 * 256) {
    int which = b / 256;       // which i_addr[0..6]
    int off = b % 256;         // which entry inside it
    diskimg_readsector(fs->dfd, in->i_addr[which], ptrs);
    return ptrs[off];
  }

  b -= 7 * 256;
  int outer = b / 256;
  int inner = b % 256;

  diskimg_readsector(fs->dfd, in->i_addr[7], ptrs);
  diskimg_readsector(fs->dfd, ptrs[outer], ptrs2);
  return ptrs2[inner];
}
```

V6 large inode: `i_addr[7]` is doubly indirect. Do not treat it as a data sector.

`file_getblock`: Logical Block To Bytes

```
int file_getblock(fs, int inum, int lbn, void *buf) {
  struct inode in;
  if (inode_iget(fs, inum, &in) < 0) return -1;

  int size = inode_getsize(&in);
  int first = lbn * DISKIMG_SECTOR_SIZE;
  if (first >= size) return 0;

  int sector = inode_indexlookup(fs, &in, lbn);
  if (sector < 0) return -1;

  if (diskimg_readsector(fs->dfd, sector, buf)
      != DISKIMG_SECTOR_SIZE) return -1;

  int remaining = size - first;
  return remaining < DISKIMG_SECTOR_SIZE
       ? remaining
       : DISKIMG_SECTOR_SIZE;
}
```

Return valid bytes. Only the final block may be short.

Directory Scan

```
int directory_findname(fs, char *name, int dirinum,
                       struct direntv6 *out) {
  struct inode dir;
  inode_iget(fs, dirinum, &dir);
  if (!(dir.i_mode & IALLOC)) return -1;
  if ((dir.i_mode & IFMT) != IFDIR) return -1;

  int size = inode_getsize(&dir);
  int nblocks = (size + 511) / 512;
  char block[512];

  for (int b = 0; b < nblocks; b++) {
    int valid = file_getblock(fs, dirinum, b, block);
    struct direntv6 *e = (struct direntv6 *) block;
    int n = valid / sizeof(struct direntv6);

    for (int i = 0; i < n; i++) {
      if (e[i].d_inumber == 0) continue;
      if (strncmp(name, e[i].d_name, 14) == 0) {
        *out = e[i];
        return 0;
      }
    }
  }
  return -1;
}
```

Use `strncmp`: a 14-char V6 name may not have `'\0'`.

Pathname Lookup

```
int pathname_lookup(fs, char *path) {
  if (path[0] != '/') return -1;
  int cur = ROOT_INUMBER;

  char copy[strlen(path)+1];
  strcpy(copy, path);          // strtok mutates string
  char *part = strtok(copy, "/");

  while (part != NULL) {
    struct direntv6 de;
    if (directory_findname(fs, part, cur, &de) < 0) {
      return -1;
    }
    cur = de.d_inumber;
    part = strtok(NULL, "/");
  }
  return cur;
}
```

`/usr/bin/ls`: root -> lookup `usr` -> lookup `bin` -> lookup `ls`.

Byte Offset To File Block

```
int off = 9000;
int lbn = off / DISKIMG_SECTOR_SIZE;
int within = off % DISKIMG_SECTOR_SIZE;

char block[512];
int n = file_getblock(fs, inum, lbn, block);
if (within < n) {
  unsigned char byte = block[within];
}
```

File offsets are logical. Inode translation turns logical block number into disk sector.

Scan All Allocated Inodes

```
int total = fs->superblock.s_isize *
            (DISKIMG_SECTOR_SIZE / sizeof(struct inode));

for (int inum = 1; inum <= total; inum++) {
  struct inode in;
  if (inode_iget(fs, inum, &in) < 0) return -1;
  if (!(in.i_mode & IALLOC)) continue;

  // check property exam asks about
}
```

Use for "find file/inode/block with property X."

Does This Inode Use Disk Block X?

```
bool uses_block(fs, struct inode *in, int target) {
  int size = inode_getsize(in);
  int nblocks = (size + 511) / 512;
  for (int lbn = 0; lbn < nblocks; lbn++) {
    int sector = inode_indexlookup(fs, in, lbn);
    if (sector == target) return true;
  }
  return false;
}
```

This checks data blocks. A separate scan is needed if the question asks about indirect metadata blocks.

Find Whether Block Is Indirect

```
for each allocated inode in:
  if (!(in.i_mode & ILARG)) continue;

  // Directly named indirect blocks:
  for (int i = 0; i < 7; i++) {
    if (in.i_addr[i] == block_num) return true;
  }

  // Indirect blocks named inside doubly-indirect block:
  uint16_t ptrs[256];
  diskimg_readsector(fs->dfd, in.i_addr[7], ptrs);
  for (int i = 0; i < 256; i++) {
    if (ptrs[i] == block_num) return true;
  }
```

`i_addr[7]` itself is the doubly-indirect block; entries inside it point at singly indirect blocks.

Hard Link Skeleton

```
int target = pathname_lookup(fs, oldpath);
inode_iget(fs, target, &tin);
if ((tin.i_mode & IFMT) == IFDIR) reject;

// split new path into parent directory + final name
int parent = pathname_lookup(fs, parent_dir(newpath));
inode_iget(fs, parent, &pin);
verify parent is directory;

// scan parent directory for empty dirent slot
slot.d_inumber = target;
copy final component into slot.d_name;
tin.i_nlink++;
```

Hard link = new directory entry to same inode. No data copy, no new file inode.

Add Directory Entry Safely

```
for each directory block:
  read block
  for each dirent e:
    if e.d_inumber == 0:
      e.d_inumber = target_inum;
      memset(e.d_name, 0, 14);
      strncpy(e.d_name, name, 14);
      write block back;
      return 0;

// no free slot: allocate/append a new directory block
```

Directory maps name -> inode number. Inode stores metadata and block pointers, not the filename.

Upgrade Small Inode To Large

```
uint16_t ptrs[256];

for (int i = 0; i < 8; i++) {
  ptrs[i] = in->i_addr[i];   // old direct data sectors
}
for (int i = 8; i < 256; i++) {
  ptrs[i] = 0;
}

for (int i = 0; i < 8; i++) in->i_addr[i] = 0;

in->i_addr[0] = new_indirect_sector;
in->i_mode |= ILARG;
```

Data blocks stay where they are; only pointer structure changes.

How Many Blocks Can It Address?

```
// lecture-style example: 4KB block, 4B block pointers
ptrs_per_indirect = 4096 / 4 = 1024
direct_capacity = 12
singly_capacity = 1024
doubly_capacity = 1024 * 1024

max_blocks = 12 + 1024 + 1024*1024
```

A doubly indirect block points to indirect blocks; those indirect blocks point to data blocks.

CS111 Code Cheatsheet: VM, Replacement, Crash Recovery

Base And Bound

```
uintptr_t translate(uintptr_t va) {
  if (va >= bound) {
    trap();                  // illegal address
  }
  return base + va;          // physical address
}
```

Message passing still works because sender traps into kernel; kernel validates/copies bytes between address spaces.

Single-Level Page Translation

```
uint64_t translate(uint64_t va, pte_t *pt) {
  uint64_t off_mask = (1ULL << PAGE_BITS) - 1;
  uint64_t offset = va & off_mask;
  uint64_t vpn = va >> PAGE_BITS;

  pte_t pte = pt[vpn];
  if (!pte.valid) page_fault();
  if (!allows(pte.perm, access)) protection_fault();

  return (pte.ppn << PAGE_BITS) | offset;
}
```

- Page size `2^k` -> offset is low `k` bits.
- Offset is unchanged in physical address.

Multilevel Index Extraction

```
// 48-bit VA, 4KB page, 8B entries
// offset = 12 bits
// entries/table = 4096/8 = 512 = 2^9
// VPN bits = 48-12 = 36 -> 4 levels

offset = va & 0xfff;
L1 = (va >> 39) & 0x1ff;
L2 = (va >> 30) & 0x1ff;
L3 = (va >> 21) & 0x1ff;
L4 = (va >> 12) & 0x1ff;
```

`0x1ff` is 9 one-bits. Each level indexes one page-map page.

Page Fault Handler Shape

```
handle_page_fault(va, access) {
  if (!legal_address(va, access)) {
    kill_process();          // segfault/protection fault
  }

  frame = find_free_frame();
  if (frame == NONE) {
    victim = choose_victim();
    if (victim.dirty) write_back(victim);
    invalidate_old_pte(victim);
    frame = victim.frame;
  }

  read_page_from_disk(va_page(va), frame);
  update_pte(va_page(va), frame, valid=true);
  restart_faulting_instruction();
}
```

Dirty victim must be written back. Clean victim can be discarded.

FIFO / LRU / Clock Trace Rules

```
FIFO:
  evict oldest page loaded into memory
  hits do not change queue order

LRU:
  evict least recently used page
  hits update recency

Clock:
  if ref bit == 1: clear it, advance hand
  if ref bit == 0: evict it
```

Common bug: treating FIFO hit like LRU hit. FIFO does not refresh on hit.

Clock Algorithm Skeleton

```
int choose_victim() {
  while (true) {
    if (frames[hand].ref == 0) {
      int victim = hand;
      hand = (hand + 1) % nframes;
      return victim;
    }
    frames[hand].ref = 0;
    hand = (hand + 1) % nframes;
  }
}
```

On memory access, set that page frame's reference bit to 1. Clock gives recently used pages a second chance.

SRPT / Priority Trace Skeleton

```
while (unfinished_jobs_exist) {
  add newly arrived jobs to ready set;
  choose ready job with smallest remaining time;
  run until next arrival or job completion;
  subtract elapsed time from remaining time;
}
```

SRPT is preemptive: a new shorter job can interrupt the currently running job.

Crash Points: Create File

```
Possible writes:
1 allocate inode / free-inode map
2 initialize inode
3 allocate data block / free-block map
4 write data block
5 update inode size/pointer
6 add directory entry
```

- Dir entry before initialized inode -> name points to garbage.
- Inode points to block marked free -> future double allocation.
- Allocated but unreachable block -> leak; usually less dangerous.

Ordered Writes: Safer Create

```
write initialized data block
write initialized inode
mark inode/block allocated
write directory entry last
```

Make reachable names point only at initialized metadata/data. For deletion, remove name before freeing inode/blocks.

Crash Recovery Answer Template

```
For each possible crash point:
1. List what reached disk.
2. Say whether metadata is consistent.
3. Say what is leaked, dangling, stale, or lost.
4. Say whether fsck/log replay can repair it.
```

Precise words score: leak = allocated but unreachable; dangling = reachable pointer to invalid/free object.

Redo Logging Skeleton

```
write_log("begin T");
write_log("new inode block");
write_log("new directory block");
write_log("new free bitmap block");
write_log("commit T");
flush_log();

install inode block to home location;
install directory block to home location;
install bitmap block to home location;
checkpoint_or_clear_log();
```

- No commit record -> ignore transaction.
- Commit present -> replay all updates.
- Replay must be idempotent because crash can happen during recovery.

Write Amplification Formula

```
erase unit utilization = U
usable fraction after GC = 1 - U
write amplification ~= 1 / (1 - U)

U=.50 -> 2x
U=.90 -> 10x
U=.99 -> 100x
```

GC must copy valid pages elsewhere before erasing. High utilization means copy lots of live pages to reclaim tiny space.

Flash Out-Of-Place Write

```
write(logical_page L, data):
  Pnew = find_free_erased_physical_page()
  program(Pnew, data)       // can change 1 bits to 0 bits
  old = map[L]
  map[L] = Pnew
  mark_garbage(old)

garbage_collect(unit):
  copy still-valid pages elsewhere
  erase entire unit          // resets bits back to 1
  add pages to free pool
```

Erased flash is all 1s, not all 0s. Programming clears selected bits from 1 to 0.

Trap-And-Simulate

```
guest executes privileged instruction, e.g. CLI
CPU traps into hypervisor
hypervisor inspects instruction
hypervisor updates virtual CPU state
guest resumes after instruction
```

Guest thinks interrupts are disabled; hypervisor only disables virtual interrupts for that virtual CPU, not real machine interrupts.

System Call Trap Shape

```
user code:
  write(fd, buf, n)

CPU:
  trap into kernel mode

kernel:
  validate fd and user pointer
  copy bytes from user memory
  ask device/filesystem to write
  return result to user mode
```

A trap is a controlled switch from less-privileged code to privileged OS/hypervisor code.

Final Code Checklist

- `execvp` arrays end with `NULL`; child exits on failure.
- Every process closes unused pipe ends; parent closes too.
- Every child is waited for or intentionally orphaned.
- Every CV wait is `while (!predicate)`, not `if`.
- All shared predicate variables use one protecting mutex.
- Exact-selection problems use waiter identity records.
- V6 directory names use bounded comparison.
- V6 code distinguishes small direct inode vs large indirect inode.
- Crash answers list writes and reason after each crash point.